

HOMOTOPY INTERVALS AND AN ASPHERICITY CRITERION

UNOFFICIAL STACKS PROJECT AI DRAFTS

ABSTRACT. We prove the sieve-cutoff contraction used in the sufficient test-category criterion of Grothendieck, including the two comma-category calculations that make its cylinder projections weak equivalences. We then prove an abstract interval-comparison lemma and the subobject-classifier criterion. All weak-equivalence and closure hypotheses are explicit. These are AI-written expository drafts from *Pursuing Stacks*, sections 31 and 37, not official Stacks text or proof-assistant-checked mathematics.

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1. TWO ELEMENTARY WEAK-EQUIVALENCE ARGUMENTS

All products and hom sets are taken in fixed suitable universes. An interval datum is an object I with two maps $e_0, e_1 : \mathbf{1} \rightarrow I$. An I -homotopy from $f_0 : F \rightarrow G$ to $f_1 : F \rightarrow G$ is a map $H : F \times I \rightarrow G$ with $H i_k = f_k$, where $i_k = (\text{id}_F, e_k) : F \rightarrow F \times I$. It is called disjoint when $\mathbf{1} \times_I \mathbf{1}$ is initial, using the two different endpoints. Disjointness is not needed in the first two lemmas.

Source-map identifiers: PSM-DEF-0035; PSM-DEF-0036.

Lemma 1.1. *Let V contain identities and satisfy two out of three. If i_0, i_1 are in V and H is an I -homotopy with $f_0 \in V$, then $H, f_1 \in V$.*

Source-map identifiers: PSM-LEM-0019.

Proof. The equality $H i_0 = f_0$ gives $H \in V$ by two out of three. The equality $f_1 = H i_1$ then gives $f_1 \in V$ by the same property. In particular,

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the endpoints lie in V whenever the projection $p : F \times I \rightarrow F$ lies in V , since $pi_k = \text{id}_F$. $\square$

Lemma 1.2. *Suppose $V = Q^{-1}(\text{Iso})$ for a functor Q . If $E \xrightarrow{i} F \xrightarrow{r} E$ satisfies $ri = \text{id}_E$ and $ir \in V$, then $i, r \in V$.*

Source-map identifiers: PSM-LEM-0020.

Proof. The endomorphism $p = Q(i)Q(r)$ satisfies $p^2 = p$ and is invertible. Multiplying by p^{-1} gives $p = \text{id}_{Q(F)}$. Together with $Q(r)Q(i) = \text{id}_{Q(E)}$, this says that $Q(i)$ and $Q(r)$ are inverse isomorphisms. The definition of V finishes the proof. $\square$

We call the conclusion of Lemma 1.2 *split-projector closure*. It is an additional assumption when V is not known to be an inverse image of isomorphisms. Two out of three alone is not being used as a substitute for it.

2. A CUTOFF HOMOTOPY DEFINED BY A SIEVE

Let A be small. Write $\text{PSh}(A) = \text{Fun}(A^{\text{op}}, \mathbf{Sets})$ and let $\text{El}_A(F)$ be its category of elements: an arrow $(a, x) \rightarrow (b, y)$ is $u : a \rightarrow b$ with $F(u)(y) = x$. For a small category B define

$$J_A(B)(a) = \text{Fun}(A/a, B),$$

where the right side is the *set* of functors. Restrictions are precomposition with the postcomposition functors between slices. The adjunction $\text{El}_A \dashv J_A$, including its action on arrows, is proved in the companion category-models module.

Proposition 2.1. *Let (I, e_0, e_1) be a disjoint interval datum in $\text{PSh}(A)$. If B has a chosen final object b_* , there is an I -homotopy from $\text{id}_{J_A(B)}$ to the constant endomorphism selecting the constant functor with value b_* .*

Source-map identifiers: PSM-PROP-0014.

Proof. For $T : A/a \rightarrow B$ and $u \in I(a)$, define a full subcategory S_u of A/a by the condition

$$(g : x \rightarrow a) \in S_u \iff I(g)(u) = e_0(x).$$

If g belongs to S_u and $k : y \rightarrow x$, naturality of e_0 gives $I(gk)(u) = e_0(y)$. Thus S_u is a sieve.

Define T_u to agree with T on S_u and to be constantly b_* on its complement. For an arrow with target in S_u , its source is also in S_u , and use T on that arrow. For an arrow with target outside S_u , use the unique map to b_* . These rules respect composition: a composite ending in S_u lies entirely in S_u ; one ending outside it has its prescribed unique image in b_* . They also respect identities, including id_{b_*} outside the sieve.

For $v : a' \rightarrow a$, the inverse image of S_u under $v_* : A/a' \rightarrow A/a$ is exactly $S_{I(v)(u)}$. Hence $(T, u) \mapsto T_u$ commutes with restrictions and defines a map $H : J_A(B) \times I \rightarrow J_A(B)$. At $u = e_0(a)$ the sieve is all of A/a ; at $u = e_1(a)$ it is empty, because disjointness is evaluated objectwise in a presheaf category. Thus the two endpoints are exactly the asserted maps. $\square$

For an arrow $x \rightarrow y$ in the slice, the construction has three cases:

positions of its endpoints	image arrow
$x, y \in S_u$	$T(x) \xrightarrow{T(x \rightarrow y)} T(y)$
$x \in S_u, y \notin S_u$	$T(x) \longrightarrow b_* \quad (\text{unique})$
$x, y \notin S_u$	$b_* \xrightarrow{\text{id}} b_*$

There is no fourth case crossing back into the sieve.

3. PRODUCTS AND A SUFFICIENT COUNT CRITERION

Fix a class W of functors between small categories. Assume that it contains isomorphisms, satisfies two out of three, and contains every right W -aspheric functor. Here C is W -aspheric when $C \rightarrow \mathbf{1}$ lies in W , and $f : C \rightarrow D$ is right W -aspheric when $(f \downarrow d)$ is W -aspheric for every $d \in D$. A presheaf is W -aspheric when its category of elements is. We call A product-aspheric if $\text{El}_A(h_a \times h_b)$ is W -aspheric for every $a, b \in A$, with $h_a = A(-, a)$. Products here are products of presheaves; A need not have products.

Source-map identifiers: PSM-DEF-0037; PSM-DEF-0038.

Proposition 3.1. *If A is product-aspheric and I is a W -aspheric presheaf, then for every $F \in \text{PSh}(A)$ the projection $\text{El}_A(F \times I) \rightarrow \text{El}_A(F)$ is right W -aspheric and belongs to W .*

Source-map identifiers: PSM-PROP-0015.

Proof. First consider $q_a : \text{El}_A(h_a \times I) \rightarrow \text{El}_A(I)$. Over (b, u) its comma category is canonically

$$(q_a \downarrow (b, u)) \cong \text{El}_A(h_a \times h_b).$$

An object on the left is an object c , a map $c \rightarrow a$, an element $t \in I(c)$, and an arrow $v : c \rightarrow b$ with $t = I(v)(u)$. The last equality determines t , leaving exactly the two arrows defining an object on the right. Morphisms on both sides are maps between their domains commuting with those two arrows, so the identification is an isomorphism, not just a bijection on objects. Consequently $q_a \in W$. Composing with $\text{El}_A(I) \rightarrow \mathbf{1}$ shows that $\text{El}_A(h_a \times I)$ is W -aspheric.

For the original projection, the comma category over $(a, x) \in \text{El}_A(F)$ is canonically $\text{El}_A(h_a \times I)$: the arrow to (a, x) forces the F -coordinate to be $F(v)(x)$, leaving $v : c \rightarrow a$ and $t \in I(c)$. The same arrow calculation verifies

the isomorphism. Every such comma category is aspheric. The assumed right-asphericity implication now gives membership in W . $\square$

Proposition 3.2. *In addition, assume that categories with final objects are W -aspheric and that W has split-projector closure. Suppose that A is W -aspheric and product-aspheric, and that it admits a disjoint interval datum whose presheaf is W -aspheric. Then every evaluation counit $\epsilon_C : \text{El}_A(J_A(C)) \rightarrow C$ belongs to W .*

Source-map identifiers: PSM-PROP-0016.

Proof. Call a presheaf map weak when its image under El_A lies in W . For a category B with final object set $F = J_A(B)$. Proposition 3.1 makes $F \times I \rightarrow F$ weak, so both endpoint sections are weak. The cutoff homotopy and Lemma 1.1 make the constant endomorphism of F weak. It factors as $F \xrightarrow{q} \mathbf{1} \xrightarrow{c} F$ with $qc = \text{id}_F$. Apply split-projector closure after El_A to get $\text{El}_A(q) \in W$. Since $\text{El}_A(\mathbf{1}) = A$ is aspheric, $\text{El}_A(F) \rightarrow \mathbf{1}$ lies in W .

For arbitrary C and $c \in C$, the counit's comma category is canonically $\text{El}_A(J_A(C/c))$. Explicitly, a functor $T : A/a \rightarrow C$ and an arrow $T(\text{id}_a) \rightarrow c$ give the functor to C/c whose structural arrow at $v : b \rightarrow a$ is $T(v) \rightarrow T(\text{id}_a) \rightarrow c$. This also identifies morphisms. Since C/c has final object id_c , the preceding paragraph makes this comma category aspheric. Thus ϵ_C is right aspheric and therefore lies in W . $\square$

Remark 3.3. The received statement assumes that W is an inverse image of isomorphisms. The proof only needs split-projector closure; the stronger assumption implies it by Lemma 1.2. No necessity claim or example of a test category is deduced here. In particular, the simplex and cube examples still require separate arguments.

4. ABSTRACT HOMOTOPY INTERVALS

Let $\mathcal{E}$ have finite products and let V contain isomorphisms, satisfy two out of three, and have split-projector closure. Say that K is universally V -aspheric if $X \times K \rightarrow X$ lies in V for every X . If there is a strict initial object and the endpoint pullback exists, a homotopy interval is a universally aspheric interval datum with disjoint endpoints.

Source-map identifiers: PSM-DEF-0044; PSM-DEF-0045.

Lemma 4.1. *Suppose I is universally V -aspheric with two global points. If F has a point $i : \mathbf{1} \rightarrow F$ and an I -homotopy from id_F to ir , where $r : F \rightarrow \mathbf{1}$, then F is universally V -aspheric. The endpoints of I need not be disjoint for this conclusion.*

Source-map identifiers: PSM-LEM-0022.

Proof. Fix X and multiply the homotopy by X , using the canonical associativity isomorphism of products. The projection from its cylinder $(X \times F) \times I$

is weak, so its endpoint sections are weak. Lemma 1.1, starting with $\text{id}_{X \times F}$, shows that $\text{id}_X \times (ir)$ is weak. This is the split projector of $X \times F \rightarrow X$ and its section $\text{id}_X \times i$. Split-projector closure makes that projection weak. This holds for every X . $\square$

Lemma 4.2. *Let I be universally V -aspheric with endpoints e_0, e_1 . Suppose L has points λ_0, λ_1 and a map $m : L \times L \rightarrow L$ satisfying*

$$m(\lambda_0, x) = x, \quad m(\lambda_1, x) = \lambda_1$$

as identities of morphisms. If $\phi : I \rightarrow L$ preserves endpoints, then L is universally V -aspheric. If its endpoints are disjoint, it is a homotopy interval. No associativity or commutativity of m is required.

Source-map identifiers: PSM-LEM-0023.

Proof. The map $H : L \times I \rightarrow L$ given by $H(x, u) = m(\phi(u), x)$ has endpoints id_L and the constant map with value λ_1 . This is a morphism formula, using the product symmetry, not an assumption that objects are sets. Apply Lemma 4.1; the final assertion is then the definition. $\square$

Proposition 4.3. *Suppose $\mathcal{E}$ has finite limits, a strict initial object 0 , and a subobject classifier Ω . It admits a homotopy interval relative to V if and only if Ω is universally V -aspheric. In that case its truth and falsehood points give a homotopy interval.*

Source-map identifiers: PSM-PROP-0028.

Proof. The classifier represents subobjects. Intersecting subobjects therefore defines a morphism $\wedge : \Omega \times \Omega \rightarrow \Omega$. The point $\top$ classifying the full subobject is a left unit; the point $\perp$ classifying the empty subobject is absorbing. Here strictness ensures that empty subobjects remain empty under pullback. By the defining classifier square for $0 \hookrightarrow \mathbf{1}$, the pullback of $\top$ and $\perp$ is 0 . Thus if Ω is universally aspheric it is a homotopy interval.

Conversely, let (I, e_0, e_1) be a homotopy interval. Every map from the terminal object is a monomorphism. The characteristic map $\chi : I \rightarrow \Omega$ of $e_0 : \mathbf{1} \hookrightarrow I$ consequently satisfies $\chi e_0 = \top$ and $\chi e_1 = \perp$: these equations classify respectively its full and empty pullbacks along the endpoints. The diagram records those two different pullbacks:

$$\begin{array}{ccc} \mathbf{1} & \xrightarrow{\text{id}} & \mathbf{1} \\ \text{id} \downarrow & & \downarrow e_0 \\ \mathbf{1} & \xrightarrow{e_0} & I \end{array} \quad \begin{array}{ccc} 0 & \longrightarrow & \mathbf{1} \\ \downarrow & & \downarrow e_0 \\ \mathbf{1} & \xrightarrow{e_1} & I. \end{array}$$

Apply Lemma 4.2 with $m = \wedge$, $\lambda_0 = \top$, and $\lambda_1 = \perp$. It makes Ω universally aspheric. $\square$

SOURCE AND DEPENDENCY NOTES

The historical source is Grothendieck, *Pursuing Stacks*. We use the edition arXiv:2111.01000v2, at section 31 (manuscript pages 39–41) and section 37 (pages 59–61). The local source IDs are not official Stacks tags. Exact source spans are recorded in `intervals-source-map.json`. The companion *Categories of Elements, Relative Nerves, and a Coint Criterion* supplies the adjunction and the prior coint criterion. This module proves a sufficient condition for that criterion, not a model structure or a comparison with spaces for arbitrary A . The historical classifier paragraph accidentally prints the first endpoint twice in its empty-intersection formula; the proof above uses the two distinct endpoints. This is a historical-source correction, not an erratum in official Stacks. The complete editable text is this file; the source archive includes build instructions and licensing notices.